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29 Boundaries of Digital Infrastructures in India

Abstract: Drawing from scholarship on infrastructure popularized in media studies, this chapter explores what happens when screen-based cultures combine with those mediated through physical apparatus and technologies. Particularly, this is done through the case of *sound trucks* or *DJ trucks* in North India and their various digital and social imbrications. Stretching the boundaries of expression and relation through and beyond digital media, the chapter works with instances demonstrating automobile cultures, masculine self-expression, toys, and sonic face-offs. Overall, the concern remains on how people relate to each other in public, how sensations and affect are mediated, and how potential for (re)action comes about. This line of thinking approaches internet and digital media not just as technical systems but as extending into and altering experiences and expression beyond devices and screens.

Keywords DJ trucks, digital infrastructure, masculinity, toy cultures, North India, street sound technologies

A lot of ink has been spent on trying to reveal and grasp the infrastructure that constitutes digital communication and culture. For instance, scholars have written about sites where data servers are located or energy concerns around mining for cryptocurrency (Hogan 2015; Cooper et al. 2019). These attempts at evincing infrastructure trace back to the materialist turn in media studies that has substantially overlapped with a similar emphasis in anthropology, urban studies, and sociology, among other disciplines.¹ There is no doubt that unearthing infrastructure is crucial to understanding the materiality of the internet and its complex interconnections with geography, geology, and resources that operate often at imperceptible scales. As a result, infrastructural approaches have done a public service by demystifying ethereal metaphors—such as platforms and cloud—used to disguise the functioning of digital computing as neutral or universal. However, when it comes to computational technologies, the discussion of infrastructure is often restricted to seeing the material behind digital media, a sort of revealing or looking underneath for substrates that can elucidate more about what appears at the surface—on the screens.

In this chapter, I try to understand how digital media intersect with and shape our physical, social, and embodied worlds. By extending perspectives on infrastructure popularized in media studies, I explore what happens when screen-based cultures combine with those mediated through physical apparatus and technologies. I demon-

¹ For a survey of cross-disciplinary work relating to infrastructure, refer to Edwards (2002) and Larkin (2013).

strate this with the example of *sound trucks* or *DJ trucks* in North India—trucks loaded with DJ equipment and speakers—and their various imbrications with digital media and social practices linked with digital media: through automobile cultures, masculine self-expression, toys, and sonic face-offs (or when two DJ trucks engage in competition). Stretching the boundaries of expression and relation through and beyond digital media, I am concerned with how people relate to each other in public, how sensations and affect are mediated, and how potential for (re)action comes about. This line of thinking approaches the internet and digital media not just as technical systems but as extending into and altering experiences and expression beyond devices and screens.

Working through the details of my site-specific case-study, I start by explaining DJ trucks and their associations with entertainment, music, and conflict. Subsequently, I outline two crucial intersections of these sound trucks with digital expression and circulation: the expression of automobiles and masculinity on audio-visual platforms and the playful mediation of toy culture and sonic face-offs. Through these instances, I understand how digital media spills into physical modes and materials of building affect and intensities. Concurrently, I put the boundaries of what constitutes infrastructure of digital media (for the larger field), and the infrastructure of violence (within the specific region of study) to the test.

DJ Trucks in Contemporary North India

On the streets of North India, automobiles often combine with loud music in public. Even as the production, consumption, and circulation of music is increasingly mediated through and for digital technologies—via phones, audio-visual and streaming platforms, and Internet-based sharing—physical amplification in public hasn't ceased to exist (Morris 2020). In fact, digital media have given an impetus to street-based sound technologies to grow and expand through cultures of knowledge sharing, competitiveness, and regional differences. For the last three years, I have been interacting with owners and operators of loudspeaker trucks—popularly called DJ trucks—in suburban and rural North India. A speaker truck is, as the name implies, a large truck retrofitted with loudspeakers favoring a bass sound that moves through the streets audibly shaking bodies and objects it encounters. These sound trucks are often owned by hobbyists, who purchase and modify these partly for investment and, partly, entertainment. The trucks, like other automobile-based cultures in the region, have become a means of expressing masculinity and caste identity by flexing the ability to capture and transform the street space.² The annoyance experienced by residents who live in and around these neighborhoods is crucial to the expression of power exerted by sound truck owners, operators, and those who rent these on festive occasions.

2 For a review of work on masculinity and geography, refer to Lohokare (2020).

During ethnographic fieldwork in the north Indian states of Haryana and Rajasthan, I observed that the common celebratory occasions when trucks were brought onto the streets included weddings, religious *yatras* (processions), births of male children, women's *sangeet* (singing) and devotional night gatherings. People I spoke to informed me that while singing and percussion drums (*dhols*) playing during these occasions has been the norm historically, sound trucks have become a more convenient and a rather popular option in the recent past. However, celebratory occasions aren't the only events when trucks come out onto the streets. More recently, speaker trucks have also become a means of stirring inter-community conflict and for the forceful expression of ethno-nationalist messaging. Journalists and scholars have noted the presence of loudspeakers in and around charged incidents that have ultimately led to violent altercations and killings, especially those targeting the Muslim minorities across the country (Prakash 2022). The use and deployment of speaker trucks to stir up violence has also been connected to a category of popular music called Hindutva Pop. This category of music, increasingly becoming popular across private and public consumption, has been identified with explicit lyrics pertaining to Islamophobia, xenophobia, and an emphatic restatement of ideas of a Hindu nation through music (Baishya 2022). Pop-music circulation, however, is just one of the tools in the digital repertoire of Hindu nationalists. It is a part of the larger digital machinery that channels platform-based affordances (such as encryption-based group messaging apps and social media bots), neighborhood group networks, and recording technologies to amplify hate speech, promote ethno-nationalist fissures, and embolden vigilante action (Nizaruddin 2021; Mukherjee 2020).

However, in public speaker truck-led events, the representational (lyrics or themes of hatred) combines with the non-representational (the affect, sensations, foreboding, etc.) to create an atmosphere of charged intensity. The modality, or technology, through which sound reaches people, especially in collective spaces, is critical because it shapes subsequent experiences, interactions, and responses. As many have argued, despite the individualization of music in the twentieth and twenty-first century with personalized listening devices, forceful beats continue to resonate communally—energizing and stirring emotions (Henriques 2014; Garcia-Mispireta 2023). DJ trucks—combining mobility and amplification—extend beyond entertainment into the realm of public life. They become tools of celebration, political mobilization, and ethno-nationalist performance, sometimes all-together at once. Through their mere presence they enable intensities that can translate into potential response and action. Concurrently, they enable assertion of dominance and power spatially through their ability to be mobile. In this sense, these technologies become part of the broader infrastructure of digital media, mediating public life through sound and influencing how space, identity, and power are negotiated. The commonly understood meaning of infrastructures as technical systems hidden behind the scenes, enabling the processes of modernity to play out, undergoes a shift here (Simone 2004). We are forced to reckon with conditions that enable certain configurations of power to come into play, embodied dynamics to shape, and sensation

and intensities to build up. Here, I am identifying uses of digital and physical media coming together to create these conditions repeatedly as ‘infrastructural.’

While sound trucks extend the infrastructure of digital media, especially as they facilitate the expression of contested identities and violence in the world, speaker trucks themselves are increasingly assembled through digital circulation of music and technical know-how, sharing of expert domain knowledge, and community-building on the internet. Digital and physical media bleed into each other seamlessly; the ruptures, if any, only become evident when one tries to see them as distinct. No avenue of life can exist without digital mediation, and hobbyist truckers of North India are no exception to it. The most evident use of digital channels here includes the sharing of modified digital music: popular sound tracks are mixed with beats suited to street-based loudspeakers. The sound mixing is carried out by a few and utilized by several DJs across the regional scene, popularized through piracy.³ Street DJ practitioners are also tied into other platform-based subcultures that include fandom around regionally popular DJs, sharing videos of truck competitions on platforms, discussing concerns around copyright and sampling, and becoming a part of trends on automobile modification that overlap with these hobbyists. In the next section I will discuss a few instances in which sound truck owners and operators leverage digital platforms to organize as a community. In observing the digital infrastructures as constitutive of embodied relations and conditions that facilitate them, I will attempt to understand and trace how distinct trends and hobby categories in esoteric corners of the internet come together to become cohesive, organized, and replicable as infrastructures.

Sound Trucks Meet Digital Platforms

Trucks, including sound trucks, are a crucial part of the economy in north India. DJs—regionally understood as a composite of truck owners, operators, and drivers—become known by their street names gaining recognition through social media profiles and content. Almost every instance when I met such DJs, they showed me their Instagram or YouTube presence with page names like DJ Raavan and DJ Fauji. The content in these pages included posts about set-up, modification to their rig, scenes from recent performances, and recordings of competitions with other DJs. During conversations, they often name-dropped other, more popular, DJs they looked up to. If I hadn’t heard of them, they would immediately pull out their social media profiles and familiarize me with their ‘idols.’ Additionally, among listeners and connoisseurs of this brand of entertainment, such as groups of young men I frequently spoke to, there was often regional consensus on popular DJs known for their stunning equipment. Equipment, including the numbers of speakers and sizes of truck, often directly translated to power and influence within the scene. ‘Set-up’—or the assembling of bass and

3 For work on piracy and music in India, refer to Athique (2008).

top (higher frequency) speakers was a driving factor for popularity, along with the ability to modify large trucks to fit heavy equipment and navigate through narrow streets. During *kawar yatra*, a pilgrimage to bring back holy water from Haridwar in the Himalayas for Shiva worship, thousands of giant sound trucks would ascend from the north Indian plains towards the mountains. This led to dreadful, albeit enjoyable to the participants, scenes on city streets and highways during the monsoon months. This was a crucial instance when a DJ's popularity was put to the test. The popularity was assessed by the market rate at which the trucks, along with their complex loud-speaker set-up, were rented out by local patrons and doubled up as a ride to the mountains for ferrying a group of male devotees. Expensive DJs subsequently amassed a larger following and became local celebrities who were then interviewed for vlogs on DJ culture. In the following section, I will outline several instances of screen-based cultures intersecting with DJ trucks. This is, by no means, an exhaustive list of such reciprocity. There are a large variety of ways street trucks and their palpable reverberations are shaped through the internet, its sharing infrastructure, and streaming and audio-visual platforms.

Automobile Cultures and Masculinity

In contemporary North India, DJ trucks exist within other forms of masculine self-expression tied to possessing, modifying, and showcasing automobiles. The cultures include expensive brands of motorcycles, SUVs, and other related accessories (Bedi 2020). During fieldwork, I observed that trucks were often just part of the fleet of vehicles that hobbyist musicians and their associates owned. The link between masculinity and automobiles is not new to the subcontinent, neither is it to other parts of the world. Specifically in North India, the link between masculine performativity, religion and the city has been understood as mapped on to the fragmented nature of the post-colonial city. As understood by Sanjay Shrivastava, the “localization of masculinity [to the neighborhood scale] allows for its reaffirmation” (2023, 842). Similarly, scholars have also linked attempts at class mobility with amassing ‘bodily capital’ in hypermasculine spaces (Baas 2020). Others have studied the creation of urban leisure architecture that includes aimless wandering or roaming as a way of exerting spatial presence and exclusion (Shannon 2022). Across these studies, masculinity emerges as grappling with a crisis brought about by growing urbanization, and the blurring of caste and class distinctions in the public’s mind. Further, capitalist consumption emerges as a means to reclaim concerns of dissipating masculinity. Within this, audio-visual platforms have facilitated a space for automobile cultures and their associated ideals of masculinity to burgeon. Across ethnographic participants’ phones, I witnessed Instagram reels and videos of leisurely driving at night posted alongside pacy rap songs, drinking booze through open sunroofs, risky off-roading on SUVs, and dangerous car chases and stunts. The audio-visual content that I witnessed ranged from fairly mundane to extraordinary.

Automobile cultures, however, often translate into explicit instances of violence in a way that DJ trucks are only a small part of the larger automobile culture in the region. Earlier in 2023, a night chase, kidnapping, and murder made headlines after a local celebrity in one of Delhi suburbs livestreamed abducting three men on Facebook. Monu Manesar, the streamer, was apparently protecting cows from being slaughtered by the three Muslim men he abducted with a captive online audience. The men's families denied Manesar's allegations. An investigative report carried out by Washington Post's India bureau, mentions repeated incidents of Manesar streaming his victories of abducting Muslim men some of whom were later found dead. This report, along with several others, mention how Manesar has been given the accolade of a 'silver button' by YouTube for surpassing 10k subscribers. Despite rights-based organizations calling for moderation of violent content, social media sites continue to platform violent content that implicitly backs the state line on such incidents. While such violence can be mediated and amplified through platforms and their affordances, it is always made visceral and forceful through its extended infrastructure. In this instance, automobiles, highways, recording technologies, and supporting sharing and knowledge networks are all part of this infrastructure.

In the rapidly urbanizing extensions of the big cities Delhi and Gurugram, automobile cultures blur into audio-visual platforms in several distinct, yet interrelated, ways: 1) platforms bring popularity and reach to divisive figures who go on to become regional celebrities through their content that primarily includes owning, modifying, and flexing cars, bikes, and trucks; 2) auto cultures enable the capturing of highways, vigilantism, and gun violence which are further echoed through music videos on YouTube, Instagram, and Facebook Reels. While the DJs I interviewed didn't indicate that they double up as vigilantes, they frequently voiced Islamophobic slurs and saw themselves as 'reclaiming' power by their forceful expression both on and off the screen. It was also not uncommon to see their associates participating in more explicit forms of automobile-led violence and control of street and highways spaces.

When it comes to DJ trucks, the expert interview format stands out as contributing to the infrastructure of such violence. For instance, a YouTuber by the name of Rahul Sharan DJ vlogs runs a channel exclusively to identify popular DJs and interview them. The series engages with questions about what motivates young men to pick up DJing (owning and operating sound trucks), how they brand their trucks or themselves, and the challenges they face in this profession. Along with operating as spaces of building associations and regional networks, these interviews also provide tips on procuring, assembling, troubleshooting, and learning to operate complex sound equipment without any professional training.

Competitive Face-Offs and Truck Toys

Part of the allure of DJ trucks is their potential to be playful, almost as much as they can be threatening. Play, as a concept, is critical to how internet-based cultures are

being increasingly understood and analyzed (Galloway 2012; Jagoda 2013). In the case of sound trucks, the potential for play operates in a couple ways especially when it concerns digital mediation. First, is the remediation of sound truck competitions. These competitions emerge from frequent ritualistic events when rival groups (coming from distinct towns or urban villages), participate in sonic face-offs. Such competitions often take place on the sidelines of religious calendar events (like *kawar yatra* discussed earlier). The rival DJs take turns to blast music from their elaborate speaker set-ups that have been assembled and placed on the bodies of large, modified trucks. The sound of the audience's cheer is used to measure the experience of sound, particularly on people's visceral sensations. I would hear people describing such competitive force in terms of how it 'dries up your spit,' 'bounces off of your chest,' or 'matches your bodily frequencies.' The competitions are fairly charged events, often resulting in brawls at the end or being forced shut by law-enforcement because of their ability to lead to large-scale conflicts (Karmakar 2024). Once these booming events get recorded on attendees' phones and shared on platforms like YouTube and Instagram, they get attached to a new set of meanings, re-narration, and recollection. For instance, when I missed such a competition in 2022, an attendee who I interviewed later on tried to piece it together for me through his embodied experience. As he narrated the thrill and fear, the videos acted as evidence, even phone speakers could barely do justice in replicating his first-hand experience of DJ sound and vibration. Sometimes, in order to facilitate evidence of the fierceness of such competitions, trucks would compete with their sound next to open bodies of water. The earth-shaking vibrations would physically bounce water and stand-in visually for the embodied experience of people.

A second way in which DJ trucks and their experience is mediated on audio-visual media concerns the use of toys. YouTube channels such as 'chatpat toy TV,' for instance, have amassed followers in millions while giving demonstrations of toys shaped, almost uncannily, like DJ trucks. The remote-controlled toy trucks mirror the features of sound trucks seen on suburban north Indian streets including campy display signs, fonts referring to old Bollywood poster graphics, and multicolored LED-based motion lighting. Among different kinds of demonstration videos, some include 'fight scenes' (similar to the competitions mentioned above) between toy trucks, embellished by outlandish, childlike commentaries. Increasingly, there seems to be a commercial market for these sound truck replicas and they have found their way onto e-commerce platforms like Amazon.in and Flipkart, among others. However, toy culture also indicates a popular fan base that exists for trucks in rural and suburban parts of north India, if not beyond. During field observations, I observed young children jury-rigging speakers from old cars or stereos and soldering them with batteries on plastic to make replica sound trucks.

The playfulness afforded through competitive face-offs and their remediations, toy and fan cultures, and their subsequent commercialization through platforms indicate the myriad ways in which sound trucks are becoming normalized and part of popular culture. The rebelliousness with which they were once associated is now mundane and accorded less seriousness. However, such popularity also indicates their pervasiveness

in popular imagination in India. The pervasiveness of practice, shared technical vocabulary, and embodied ways of skilling and learning both of performers/operators and listeners make the DJ truck phenomenon infrastructures. These learned and practiced skills come in handy when they are called upon, for instance in the case of ethno-religious conflict.

Among the anthropologists of sonic phenomenon and media—particularly Brian Larkin (in his work on public loudspeakers in Nigeria) and Patrick Eisenlohr (in his work on sound, mobility, and belief in Malaysia and India)—there is consensus on focusing on the materiality of the sonic apparatus in an attempt to grasp charged affect and sensations. Across these and other such works, media infrastructures have been seen as fundamental to shaping often inarticulable ideals of faith, piety, and belief. Within the analysis of DJ trucks, sound is mediated through trucks, loudspeakers, and bodies that surround them but also through associated practices—like competitions, toy videos, and DJ interviews—that are digitally mediated. It is important to see these technological practices and mediations within a spectrum as opposed to distinct media forms.

Conclusion

So far, I have tried to argue that digital media is materially instantiated, extending its infrastructure into physical spaces. The constitutive parts of digital media often get restricted to technical elements such as servers, electrification, minerals, or interfaces (devices or platforms that allow sensory interaction). However, in global majority contexts, a range of materials that channel information beyond digital circulations will need to be thought of as infrastructure of digital media. I have given the instance of digital music, DJ trucks, and associated practices in this regard. These extended media technologies, however, cannot be comprehensively understood, as I have demonstrated, without seeing how they are digitally co-constituted and imbricated.

Media and internet studies has been, fairly, accused of approaching ‘infrastructure’ with vague and confusing boundaries across the field (Hesmondhalgh 2021). And yet, I believe that the rhetorical value of the term, brought through frequent usage, has the potential to draw attention towards anachronistically and spatio-temporally blended media practices that are common and frequent among the global majority. While materials, technologies, and infrastructure can illuminate complex identities that are being disproportionately studied as digital phenomena (including Hindutva discussed above), emerging formats and sub-cultures such as the ones I discussed earlier, can further complicate the discussion. They can give a more holistic picture towards a range of action: from mundane frictions between communities to unimaginable scales of violence.

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